



University
of Glasgow

Power Electronics

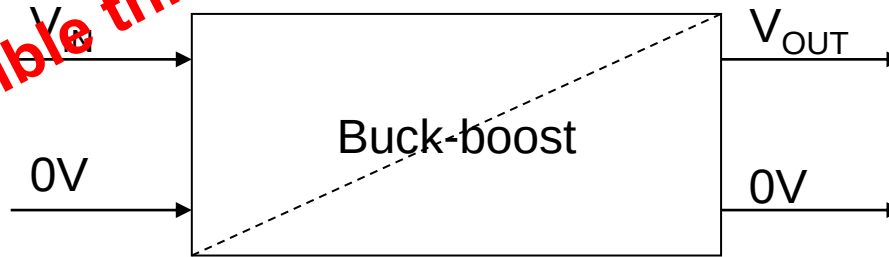
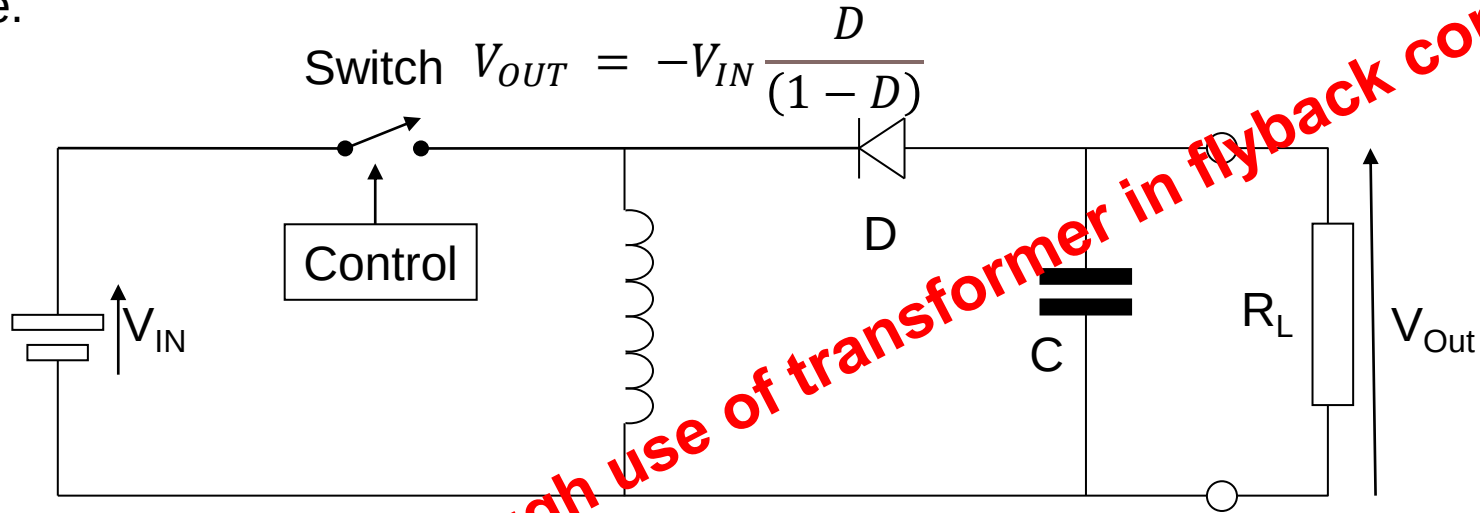
Flyback Converter and Snubber Circuit (缓冲电路)

Please read section 10-4-2 of Chapter 10
And Chapter 27 in the textbook

Flyback Converter

The buck-boost converter ($V_{OUT} < > V_{IN}$).

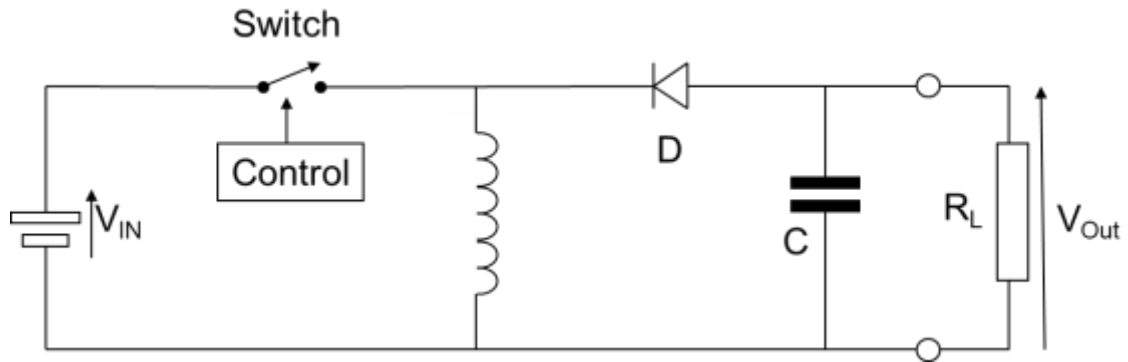
In the last lecture we saw that the Buck-Boost converter is able to produce output voltages that are higher or lower than the input voltage depending upon the duty cycle.



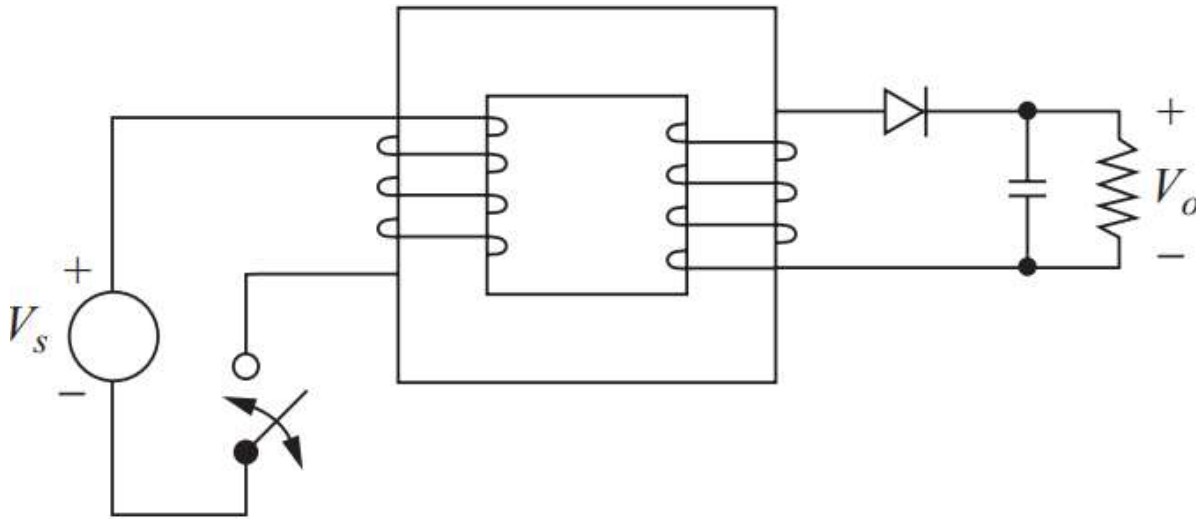
In the circuit above, the input and output are not electrically isolated. The question is, can we provide electrical isolation between input and output?

The Flyback Converter ($V_{OUT} < > V_{IN}$).

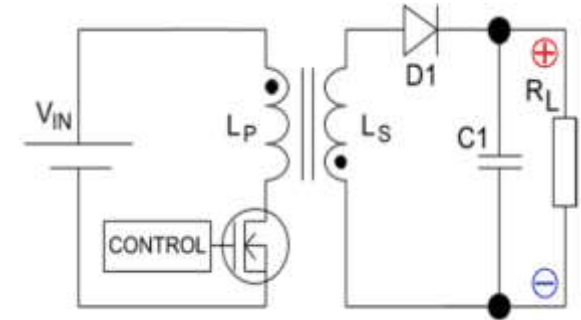
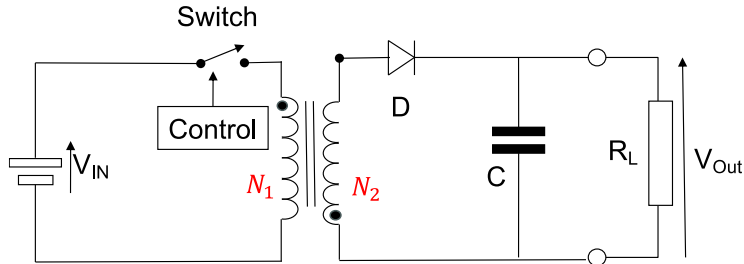
Flyback converters are derived from the buck-boost converters.



$$V_{OUT} = -V_{IN} \frac{D}{(1 - D)}$$



The Flyback Converter ($V_{OUT} <> V_{IN}$).



- The basic operation of the flyback converter is similar to that of the buck-boost converter.
- By placing a second winding on the inductor, it is possible to achieve electrical isolation. Hence, a transformer is placed with N_1 turns with inductance (L_p) on primary winding and N_2 turns with inductance (L_s) are placed on secondary winding
- When switch is closed, the current is increasing linearly in the physical primary winding, and the inductor core **flux** increases linearly from its initial value. However, due to the polarity of the secondary winding of transformer no current flows in secondary winding.
- As soon the switch is open, the voltage across inductor on primary side reverses, and due to opposite polarity of winding in secondary, diodes turns on and current flow due to stored energy in the core.
- Compared to buck-boost converter, apart from duty cycle (D), turn ratio ($\frac{N_2}{N_1}$) is also included in the relation between input and output.

Boost converter Input/Output Voltage Relation

The integral of inductor voltage is equal to the change in the inductor flux linkage. The net change of flux linking the inductor over one time period of repetition is zero, which is a necessary condition for steady-state operation.

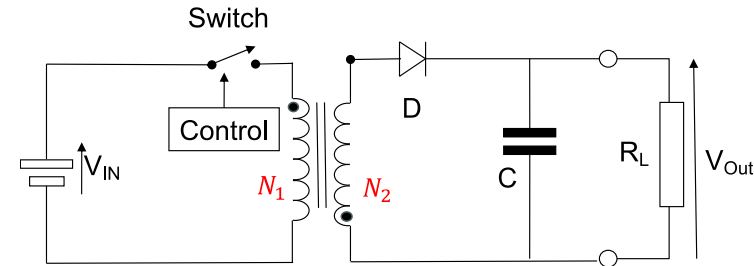
$$\int_0^T V_{L,primary} dt = 0$$
$$\int_0^{t_{on}} V_{IN} dt + \int_{t_{on}}^T -V_{OUT} \frac{N_1}{N_2} dt = 0$$

$$V_{IN} t_{on} - V_{OUT} \frac{N_1}{N_2} (T - t_{on}) = 0$$

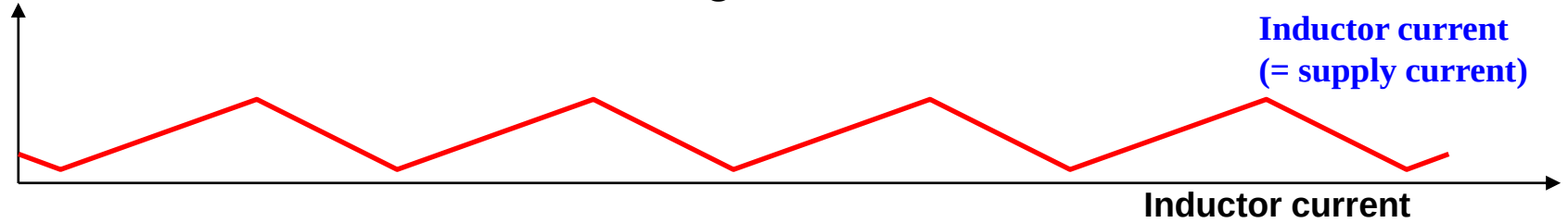
$$V_{IN} DT = V_{OUT} \frac{N_1}{N_2} (T - DT)$$

$$V_{IN} \frac{N_2}{N_1} D = V_{OUT} (1 - D)$$

$$V_{OUT} = V_{IN} \frac{D}{1 - D} \times \frac{N_2}{N_1}$$



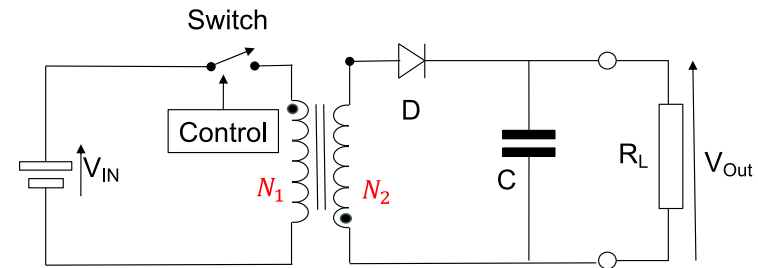
Boost converter Inductor Current Change



Computing the change in current using following expression:

$$v_L = L \frac{di_L}{dt} \Rightarrow \frac{di_L}{dt} = \frac{v_L}{L}$$

When switch is closed:



$$\frac{(\Delta i_L)_{switch\ closed}}{t_{on}} = \frac{V_{IN}}{L_p} \Rightarrow (\Delta i_L)_{switch\ closed} = \frac{V_{IN}}{L_p} \times t_{on}$$

$$\Rightarrow (\Delta i_L)_{switch\ closed} = \frac{V_{IN}}{L_p} \times DT$$

When switch is open:

$$\frac{(\Delta i_L)_{switch\ open}}{T - t_{on}} = -\frac{V_{OUT} \frac{N_1}{N_2}}{L_p} \Rightarrow (\Delta i_L)_{switch\ open} = -V_{OUT} \frac{N_1}{N_2} \frac{1}{L_p} \times (T - t_{on})$$

$$\Rightarrow (\Delta i_L)_{switch\ open} = -V_{OUT} \frac{N_1}{N_2} \frac{1}{L_p} \times T(1 - D)$$

Minimum and Maximum Inductor Current

Assuming lossless system

$$P_{IN} = P_{OUT} \Rightarrow V_{IN} I_{IN} = V_{OUT} I_{OUT} \Rightarrow V_{IN} I_{Lp,avg} D = V_{OUT} \frac{V_{OUT}}{R}$$

$$V_{IN} I_{Lp,avg} D = \frac{V_{OUT}^2}{R} \Rightarrow I_{Lp,avg} = \frac{(V_{OUT})^2}{V_{IN} D} \times \frac{1}{R}$$

$$I_{Lp,avg} = \left(V_{IN} \frac{D}{(1-D)} \frac{N_2}{N_1} \right)^2 \times \frac{1}{V_{IN} D R}$$

$$I_{Lp,avg} = \frac{V_{IN} D}{(1-D)^2 R} \left(\frac{N_2}{N_1} \right)^2$$

$$I_{Lp,max} = I_{Lp,avg} + \frac{(\Delta i_{Lp})_{switch\ closed}}{2}$$

$$I_{Lp,min} = I_{Lp,avg} - \frac{(\Delta i_{Lp})_{switch\ closed}}{2}$$

Output Ripple (same as Buck-Boost Converter)

$$\frac{\Delta V_{OUT}}{V_{OUT}} = \frac{DT}{RC}$$

Snubber Circuit

Protection of Switching Devices and Circuits

- The function of a snubber circuit is to reduce the electrical stresses placed on a device during switching by a power electronics converter to levels that are within the electrical ratings of the device.
- It must be emphasized that snubbers are not a fundamental part of a power electronic converter circuit. The snubber circuit is an addition to the basic converter, which is added to reduce the stresses on an electrical component, usually a power semiconductor device.

Snubber Circuits for Hard-Switched Converters

Function of Snubber Circuit

- Limiting device voltages during turn-off transients
- Limiting device currents during turn-on transients
- Limiting the rate-of-rise (di/dt) of currents through the semiconductor device at device turn-on
- Limiting the rate-of-rise (dv/dt) of voltages across the semiconductor device at device turn-off
- Shaping the switching trajectory of the device as it turns on/off

From the circuit topology perspective, there are three broad classes of snubber circuits

Unpolarized series **R-C** snubbers:

- Used to **protect diodes and thyristors**
- Used to Mitigate **Ringing**

Polarized **R-C** snubbers:

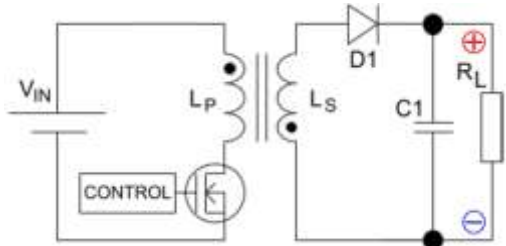
- Used as **turn-off** snubbers to shape the turn-off switching trajectory of controlled switches. Limit **dv/dt** during device turn-off
- Used as **overvoltage** snubbers to clamp voltages applied to controlled switches to safe values;

Polarized **L-R** snubbers

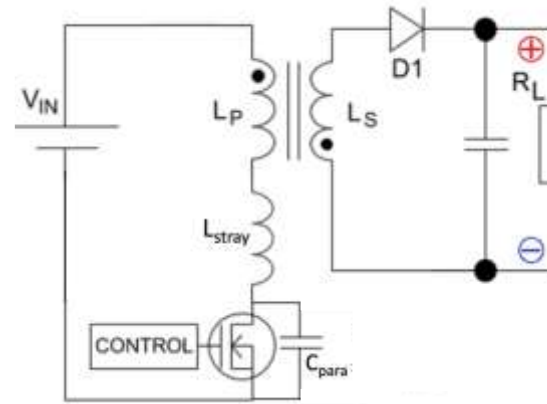
- Used as **turn-on** snubbers to shape the turn-on switching trajectory of controlled switches;
- Limit **di/dt** during device turn-on

Ringling Effect.

Real electronic switches have a “**parasitic**” capacitance between their terminals, and circuit board tracks introduce **stray** inductance between components.



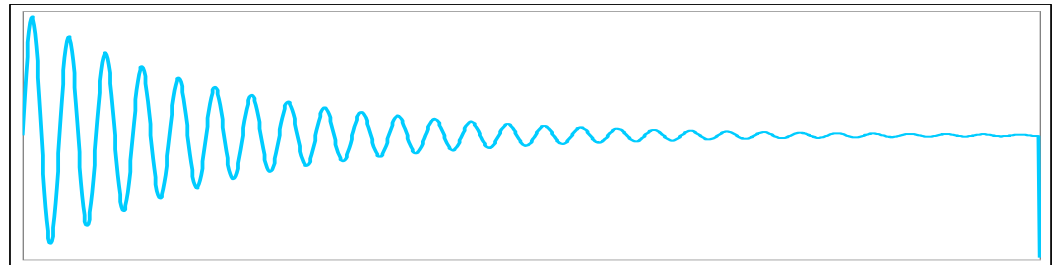
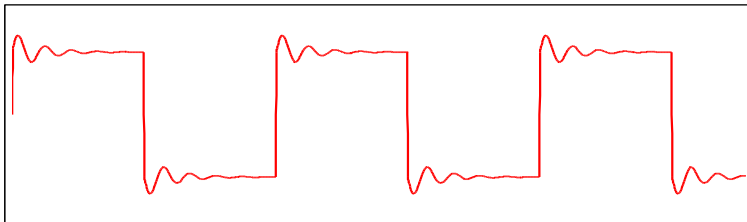
Ideal Flyback Converter



Actual Flyback Converter

At switching, this capacitance rings with the inductance at the undamped resonant frequency:

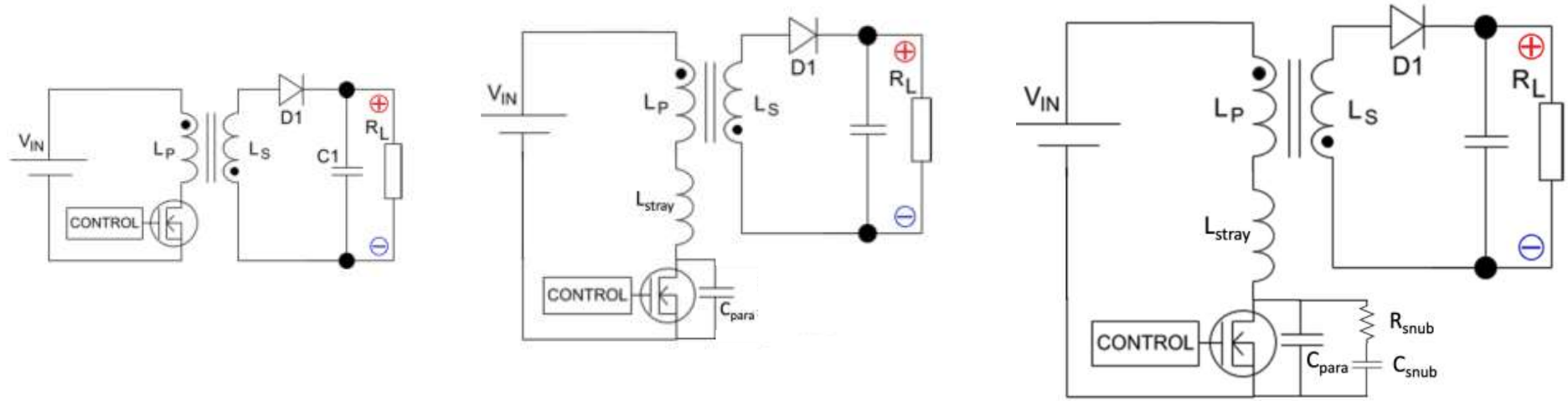
$$f_o = \frac{1}{2\pi\sqrt{LC}}$$



In these “**parasitic**” circuits there is usually very low resistance to dissipate the power in the oscillations and they can last for a considerable number of cycles.

Ringing in Flyback Converter and Series RC Snubber Circuit

Snubbers are used extensively to prevent “ringing” when a circuit is **switched on or off**. Consider, the flyback converter shown below:



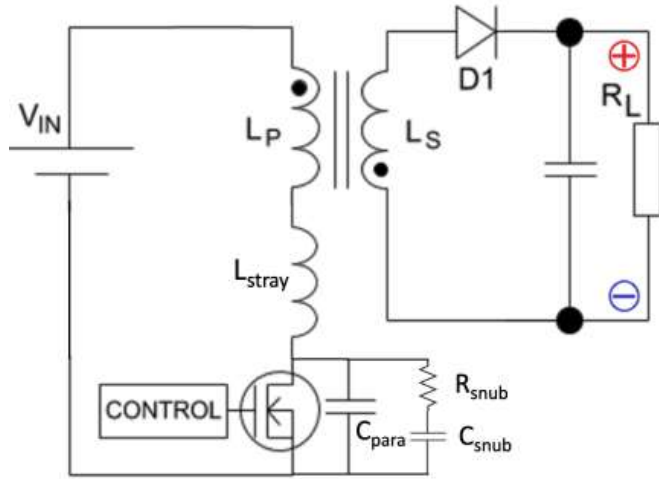
The **RC snubber** will, if properly designed, **damp the ringing and limit the overshoot** at switching. *Optimum performance is obtained when the snubber capacitance is 3X the parasitic capacitance and the resistor value equals the characteristic impedance of the un-modified resonant circuit.*

The characteristic impedance is given by

$$R_{snub} = \sqrt{\frac{L_{stray}}{C_{para}}}$$

Example.

The switch in an SMPS has a parasitic capacitance of 330pF and the stray inductance due to circuit board layout is 770nH. What is the undamped ringing frequency? An RC snubber is to be used to reduce the power dissipation in the switch and to control the ringing. What are the optimum values for R & C?

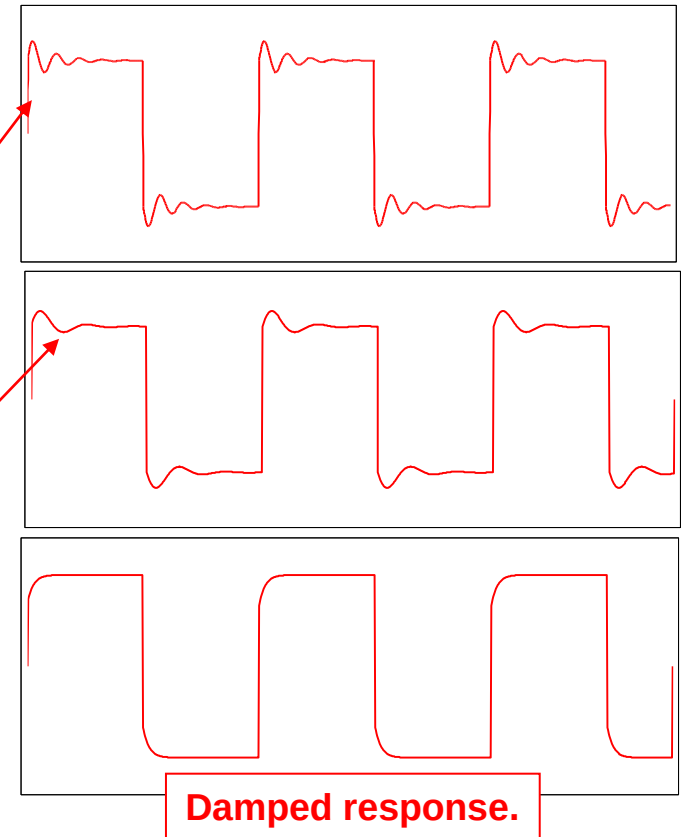


$$f_o = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{770 \times 10^{-9} \times 330 \times 10^{-12}}} \approx 10\text{MHz}$$

$$R_{snub} = \sqrt{\frac{L_{stray}}{C_{para}}} = \sqrt{\frac{770 \times 10^{-9}}{330 \times 10^{-12}}} \approx 48\Omega$$

$$C_{snub} = 3 \times C_{para} = 990\text{ pF} \Rightarrow C_{Total} = 1320\text{ pF}$$

$$f_{new} = \frac{1}{2\pi\sqrt{LC_{Total}}} \approx 5\text{MHz}$$



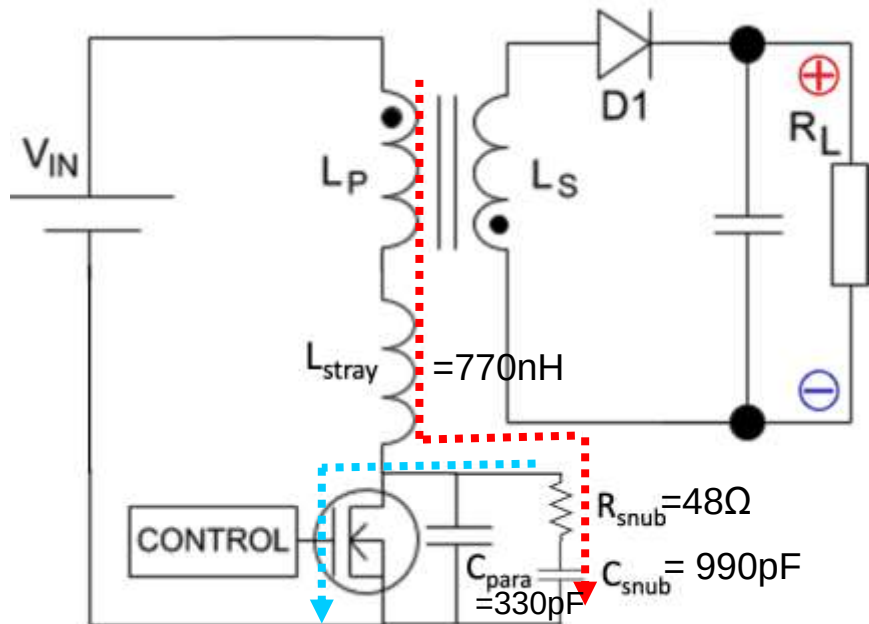
Example (contd)

If the switching frequency is 100KHz and the maximum voltage across the snubber capacitor is 100V, estimate the power that is dissipated in the snubber resistor.

Energy in a capacitor is $E = \frac{1}{2} CV^2$

When the switch is open, the capacitor **charges** to 100V via the resistor. When the switch closes, the capacitor is **discharged** via the resistor. Hence the energy dissipated per cycle is $2 \times \frac{1}{2} CV^2$ and the energy dissipated per second is $f CV^2$ (W).

$$P = f CV^2 = 10^5 \times 990\text{pF} \times 100^2 = 0.990\text{W}$$



Caveat: This works for

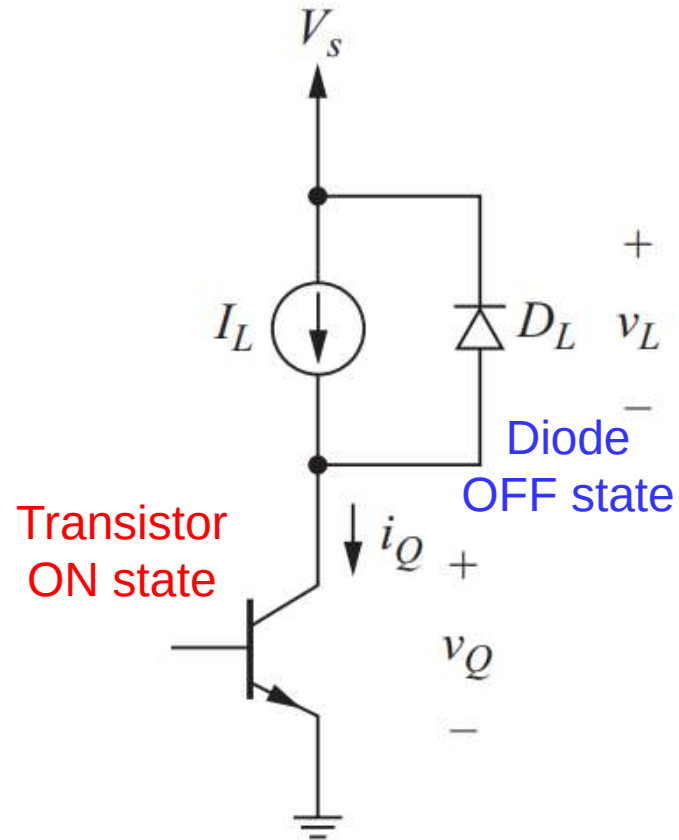
$$\tau_{Osc} \ll T_{Switch}$$

$$\tau = RC \Rightarrow \tau_{Osc} = 47.5\text{nS}$$

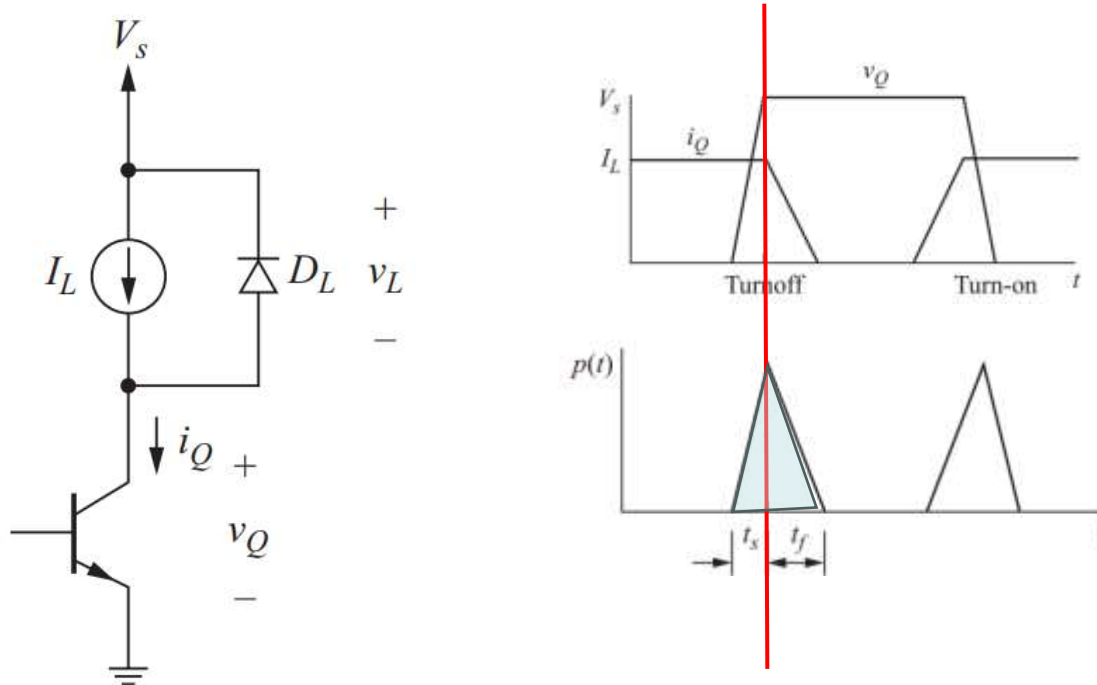
$$T_{Switch} = 1/F = 10\mu\text{S}, \text{ so OK.}$$

Note that the current from the snubber flows through the switch when the switch is closed.

Polarized R-C snubber – Turn-off Snubber

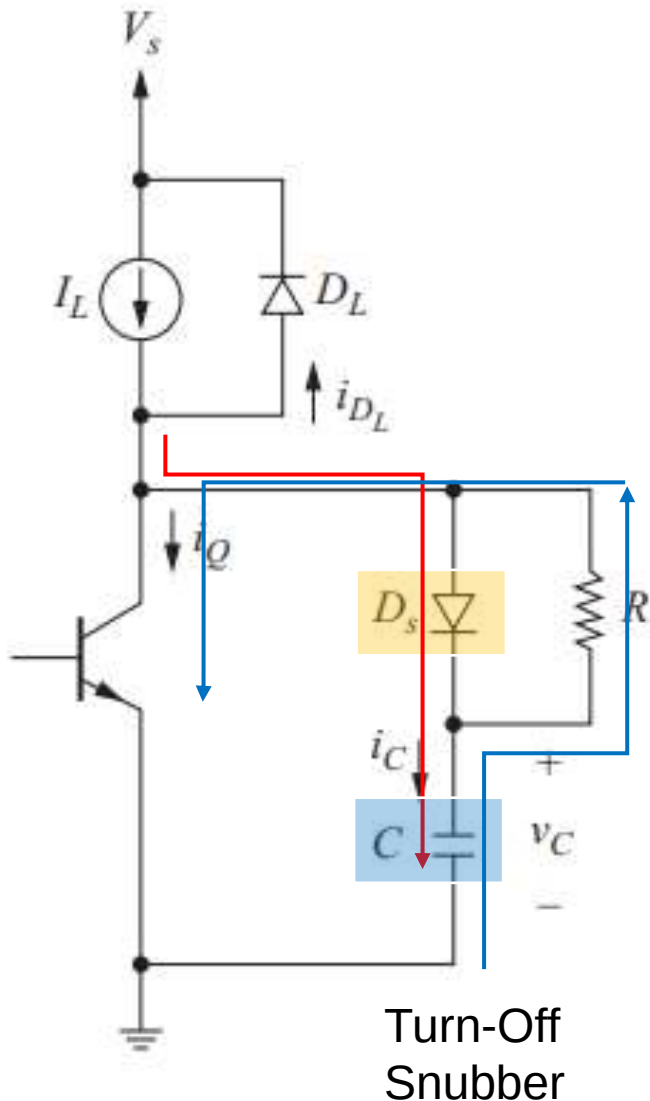


Polarized R-C snubber – Turn-off Snubber



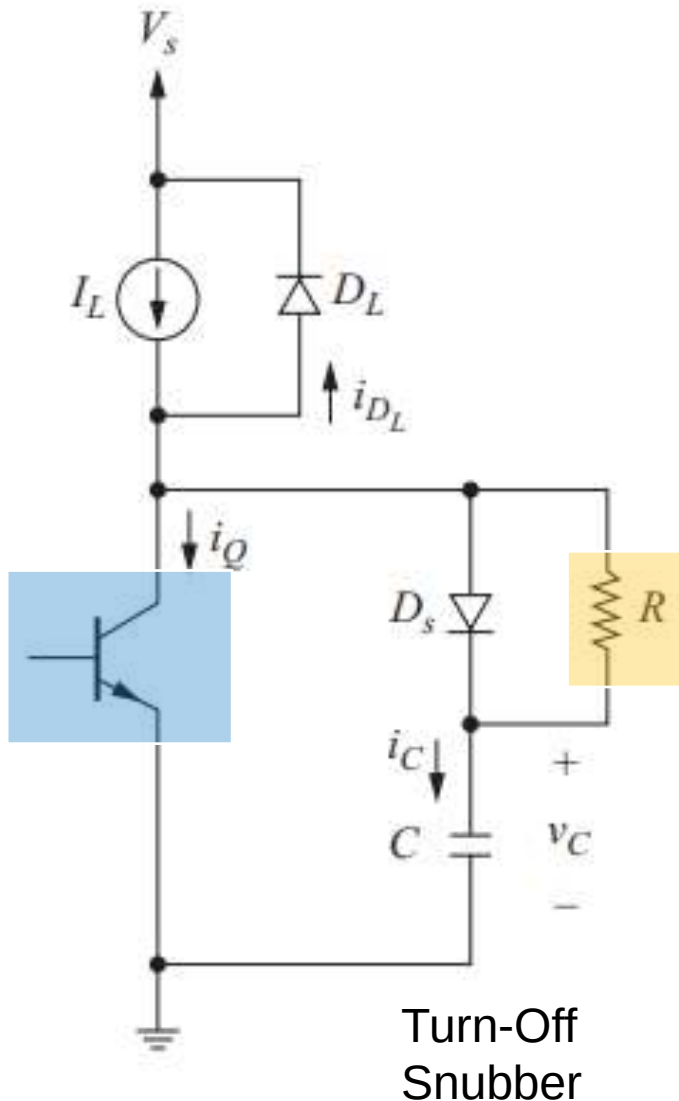
- Transistor turns off, the diode remains reverse-biased until the transistor voltage v_Q increases to the source voltage V_s .
- After the transistor voltage reaches V_s , and the transistor current decreases to zero.
- As a result, there is a point during turnoff when the transistor voltage and current are high simultaneously, resulting in a triangularly shaped instantaneous power waveform $p(t)$.

Polarized R-C snubber – Turn-off Snubber



- The snubber provides another path for load current during turnoff.
- As the transistor and the voltage across it is increasing, the snubber diode D_s becomes forward biased and the capacitor begins to charge.
- The rate of change of transistor voltage is reduced by the capacitor, delaying its voltage transition from low to high.
- The capacitor charges to the final off-state voltage across the transistor and remains charged while the transistor is off.
- When the transistor turns on, the capacitor discharges through the snubber resistor and transistor.

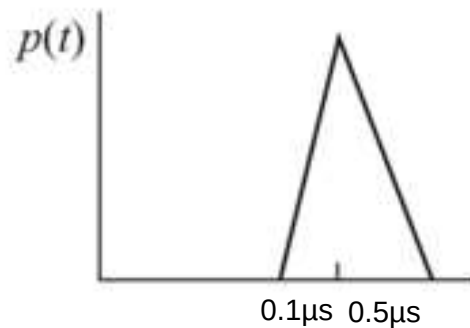
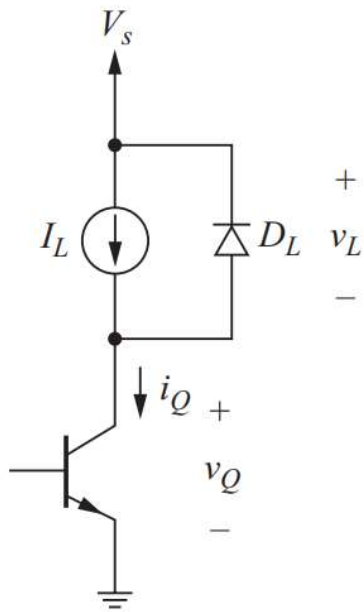
Polarized R-C snubber – Turn-off Snubber



- All the capacitor energy is dissipated in the resistor, which is easier to cool than the transistor.
- No additional energy dissipation occurs in the transistor due to the turn-off snubber.
- The peak current the transistor must conduct is not increased due to the turn-off snubber when it turns-on.

Turn off Snubber - Problem Solving

The converter and snubber in Fig. has V_s 100 V and I_L 5 A. The switching frequency is 100 kHz with a duty ratio of 50 percent, and the transistor turns off in $0.5\mu\text{s}$. **(a)** Determine the turnoff losses without a snubber if the transistor voltage reaches V_s in $0.1\mu\text{s}$. **(b)** Determine the transistor turnoff losses and the resistor power with the snubber added if 12.5 nF capacitor is used.



Solution: (a)

Transistor voltage reaches 100 V while the current is still at 5 A, resulting in a peak instantaneous power of $(100\text{ V})(5\text{ A}) = 500\text{ W}$.

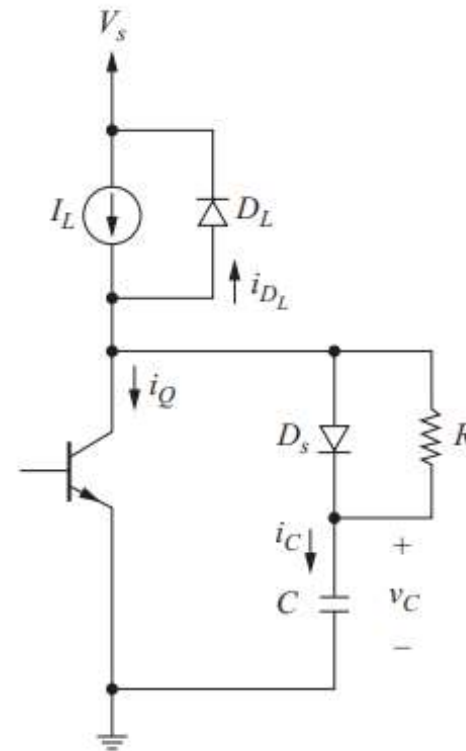
The base of the power triangle is $0.6\mu\text{s}$, making the energy = $0.5(500\text{ W})(0.6\mu\text{s}) = 150\mu\text{J}$.

Turn off Snubber - Problem Solving

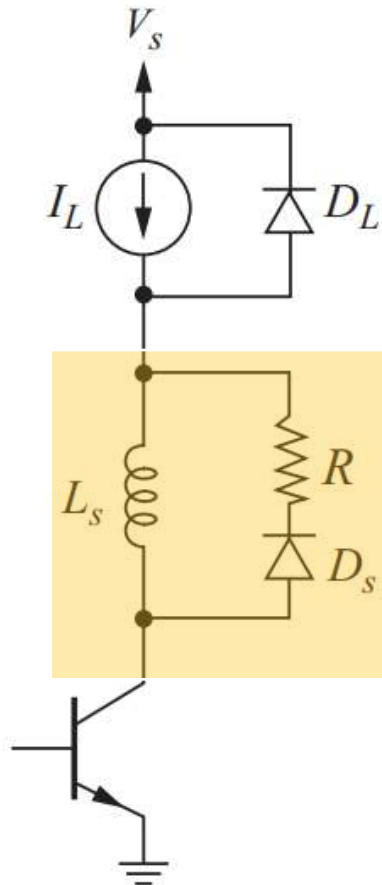
The converter and snubber has V_s 100 V and I_L 5 A. The switching frequency is 100 kHz with a duty ratio of 50 percent, and the transistor turns off in $0.5\mu\text{s}$. **(a)** Determine the turnoff losses without a snubber if the transistor voltage reaches V_s in $0.1\mu\text{s}$. **(b)** Determine the transistor turnoff losses and the resistor power with the snubber added if 12.5 nF capacitor is used.

Solution: (b)

$$P_R = \frac{1}{2} C V_s^2 f = \frac{0.0125(10^{-6})(100^2)(100,000)}{2} = 6.25 \text{ W}$$



Polarized R-L snubber – Turn-on Snubber



Turn - ON
Snubber

- Turn-on snubbers protect the device from simultaneously high voltage and current during turn-on.
- As with the turn-off snubber, the purpose of the turn-on snubber is to modify the voltage-current waveforms to reduce power loss.
- An inductor in series with the transistor slows the rate of current rise and can reduce the overlap of high current and high voltage.
- The snubber diode is off during turn-on. During turnoff, the energy stored in the snubber inductor is dissipated in the resistor.

Example

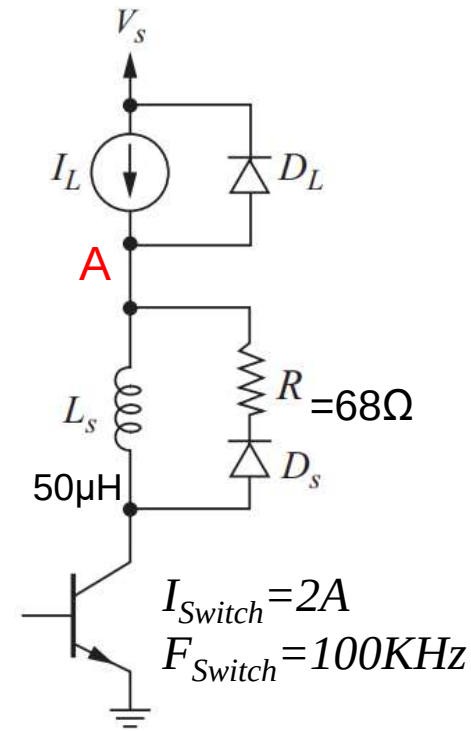
In the circuit shown, the switch operates at 100KHz. The supply voltage is **200V**. If the peak current in the switch is **2 amps**, what power is dissipated in the resistor when switch is open?

What is the rate of rise of current limited to if the voltage at **A** is 200V when the switch is closed?

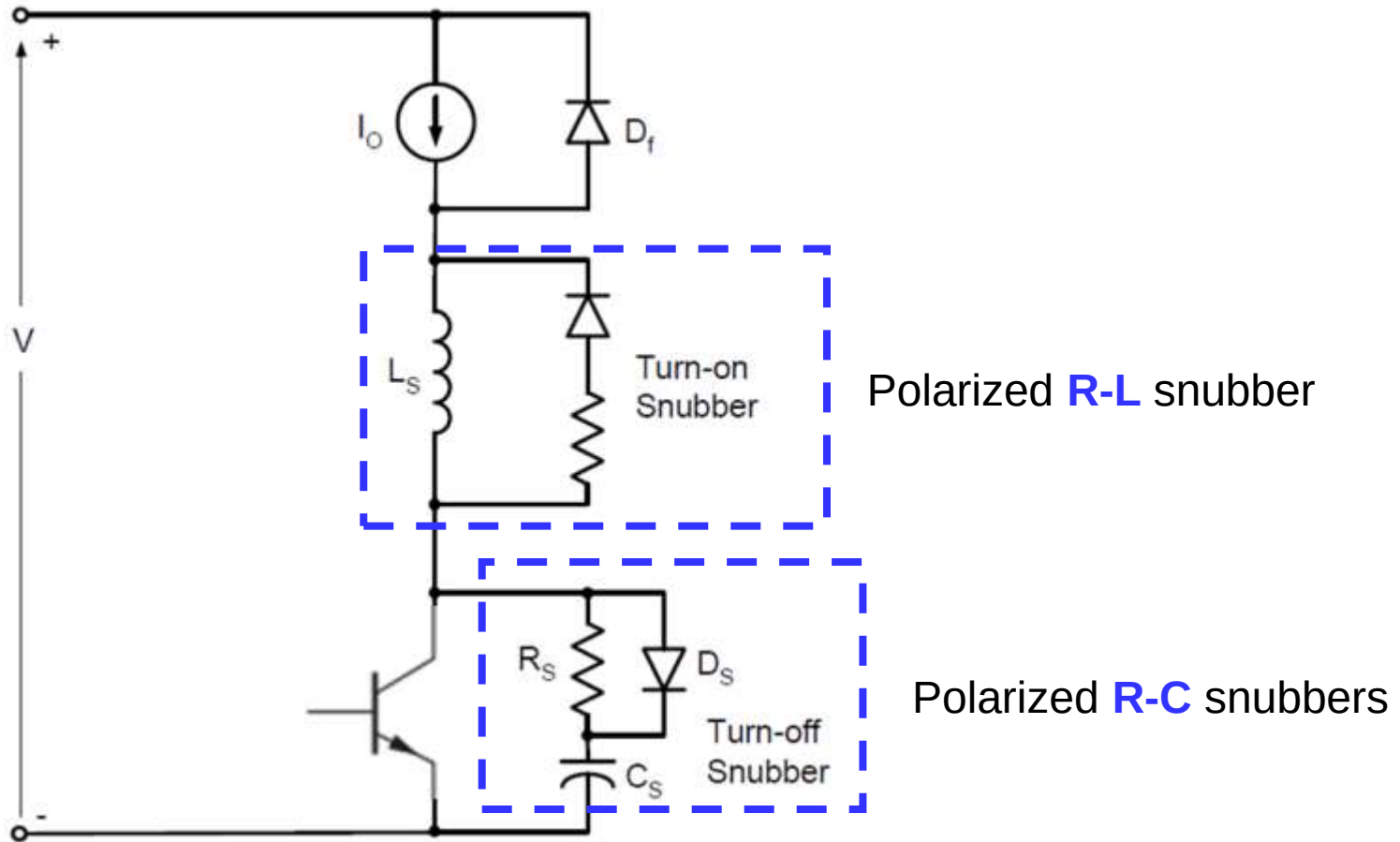
$$\text{Inductor Energy} = \frac{1}{2}LI^2 = \frac{1}{2} \times 50\mu H \times 2 \times 2 = 100\mu J$$

$$\text{Power} = \text{Inductor Energy} \times F_{\text{switch}} = 100\mu J \times 100 \text{ KHz} = 10 \text{ W}$$

$$V = L \frac{di}{dt} \Rightarrow \frac{V}{L} = \frac{di}{dt} = \frac{200}{50\mu H} = 4A/\mu\text{sec}$$

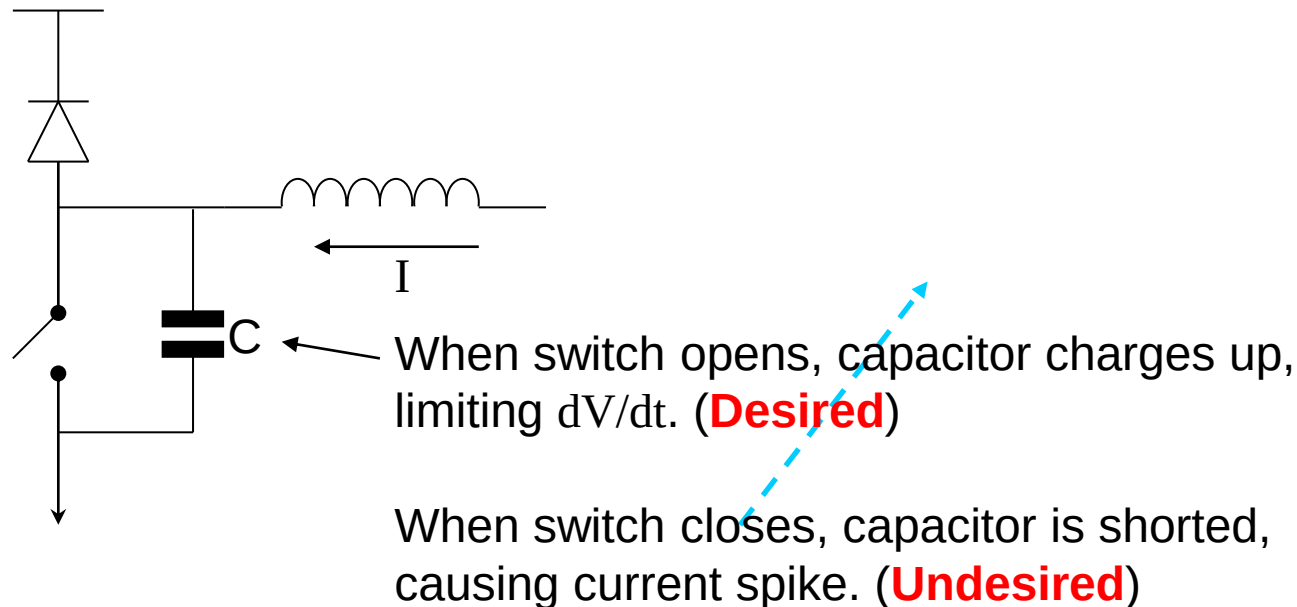


Summary of Snubber Circuit



There are two key points to note about snubber circuit design:

1. There is always **a degree of experimentation** needed to tune a snubber's operation. It is impossible to precisely design the circuit (although the desired function can be predicted).
2. The final circuit will probably be a **compromise**. A capacitive snubber that controls a voltage spike at turn-off may cause a current spike at turn-on. Similarly, controlling a current spike on one edge may cause a voltage spike on the other.



What we learned?

- Snubber Circuits
 - Why? -> functions
 - Where? -> protection
 - How? -> configurations
- Classes of Snubber Circuits
- Types of Snubber circuits
 - Series RC to mitigate ringing
 - Voltage snubbers (turn off)
 - Current snubbers (turn on)